

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

Reference: arxiv.org/abs/2504.04715

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

The Core Threat:

Black-box APIs mean users have no guarantee the advertised model (e.g., Llama-3-70B) is actually being served. Providers can covertly substitute cheaper models to reduce GPU costs.

Hypothesis Testing Framework:

Can an auditor distinguish the specified model's distribution $P_{\text{spec}}(y|x)$ from an alternative $P_{\text{actual}}(y|x)$?

Four Types of Covert Substitution:

1. Smaller Models
(e.g., 8B instead of 70B)

2. Quantized Variants
(e.g., FP8/INT8 vs FP16)

3. Fine-tuned/Updated
(Cheaper inference)

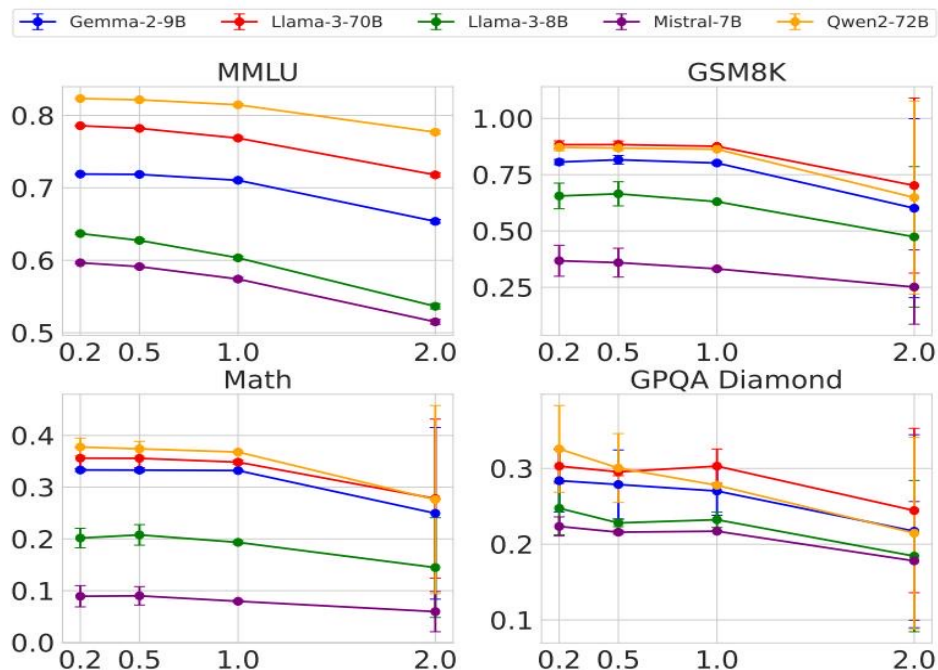
4. Different Families
(Completely different arch)

FP8 Quantization Preserves Most Accuracy

Performance drops are often within error bars, making substitution hard to detect.

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 $\pm$ 0.18	61.14 $\pm$ 3.47	20.65 $\pm$ 3.43	22.62 $\pm$ 0.22
Llama-3-8B-Instruct-FP8	62.43 $\pm$ 0.26	60.90 $\pm$ 4.10	14.91 $\pm$ 2.49	20.14 $\pm$ 0.24
Llama-3-70B-Instruct	78.05 $\pm$ 0.08	88.06 $\pm$ 1.44	35.69 $\pm$ 1.33	29.60 $\pm$ 0.51
Llama-3-70B-Instruct-FP8	77.88 $\pm$ 0.13	87.35 $\pm$ 1.24	35.75 $\pm$ 1.16	33.12 $\pm$ 0.30
Gemma-2-9b-it	71.86 $\pm$ 0.08	81.80 $\pm$ 1.35	33.41 $\pm$ 0.28	28.64 $\pm$ 2.97
Gemma-2-9b-it-FP8	71.92 $\pm$ 0.11	79.41 $\pm$ 1.14	32.53 $\pm$ 0.34	27.81 $\pm$ 3.22
Qwen2-72B-Instruct	82.18 $\pm$ 0.08	86.72 $\pm$ 1.00	37.39 $\pm$ 1.41	29.93 $\pm$ 2.71
Qwen2-72B-Instruct-FP8	81.98 $\pm$ 0.08	86.82 $\pm$ 0.97	37.67 $\pm$ 1.39	31.08 $\pm$ 1.96
Mistral-7B-Instruct-v0.3	59.15 $\pm$ 0.10	35.90 $\pm$ 4.54	8.94 $\pm$ 1.24	21.60 $\pm$ 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 $\pm$ 0.13	32.20 $\pm$ 4.02	7.68 $\pm$ 1.23	22.72 $\pm$ 0.19

Higher Temperature Amplifies Performance Gaps but Increases Noise



Temperature Scaling Effects:

- Higher temperatures (e.g., 2.0) degrade performance across all models.
- The performance gap between the original model and the substitute narrows or becomes obscured by noise.
- **Conclusion: Modifying temperature does not provide a reliable statistical signal for substitution detection.**

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

The Fingerprinting Premise:

Comparing token-level log probabilities between a local reference model and the API.

The Production Reality:

- Same model (e.g., Llama-3-8B) yields different log-probs across frameworks (transformers vs vLLM).
- Hardware changes (A100 vs H100) introduce variance.
- **Result: Inherent variation mimics substitution, rendering fingerprinting useless.**

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

Advertised Model
Probability: $(1 - p)$

+

Substitute Model
Probability: p

Adversarial Countermeasure: Randomized Substitution

Providers can route only a fraction (p) of queries to the cheaper model. As p approaches 0, the mixed distribution tightly converges to the specified model's distribution. The number of queries required for statistical detection grows exponentially, making audits financially impossible.

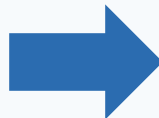
Software-Only Auditing Methods Are Fundamentally Unreliable

Method Type	Weakness / Failure Mode	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Fails against subtle substitutions (like FP8) because output distributions remain statistically indistinguishable without massive samples.	Requires impractically large query budgets; financially unfeasible for continuous auditing.
Log-Probability Fingerprinting	Defeated by production-level inference nondeterminism. Software versions (vLLM, Transformers) and hardware (A100, H100) shift log-probs natively.	False positives are guaranteed; legitimate scaling across diverse hardware looks exactly like a substitution.
Adversarial Countermeasures	Providers can actively evade audits using benchmark detection (routing known prompts to real models) or randomized low-rate substitution.	Auditors can never trust the samples they receive if the API maliciously filters or mixes traffic.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

Software-Only Auditing

- ✗ Probabilistic
- ✗ Query-prohibitive
- ✗ Vulnerable to evasion
- ✗ Nondeterminism issues



Hardware Attestation (TEEs)

- ✓ Provable Integrity
- ✓ Cryptographic Guarantees
- ✓ Modest Overhead
- ✓ Deployable Today (NVIDIA H100)

TEEs guarantee that the correct model weights are loaded and the intended inference code is running, closing the verification gap practically without requiring ZKPs.

Key Takeaways: Close the Verification Gap with Hardware Attestation

1. Model substitution is economically motivated and practically undetectable via software.
2. FP8 quantization barely affects benchmarks, masking the substitution.
3. Inference nondeterminism kills log-probability fingerprinting methods.
4. Randomized substitution and benchmark evasion defeat statistical tests.
5. Trusted Execution Environments (TEEs) are the only currently viable path to verifiable LLM APIs.

Stop guessing. Demand cryptographic attestation.